

JEE MAIN REVISION MASTER: COORDINATION COMPOUNDS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 319 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Crystal Field Theory (CFT) & Isomerism	176 PYQs	CFT & Isomerism
Very High	IUPAC Nomenclature & Werner's Coordination Theory	164 PYQs	Coordination Basics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Crystal Field Theory (CFT) & Isomerism	$\Delta_t = \frac{4}{9}\Delta_o$, $\Delta_{CFSE}(\text{oct}) = [-0.4 n(t_{2g}) + 0.6 n(e_g)]\Delta_o + n P$, $\mu = \sqrt{n(n+2)} \text{ BM}$
IUPAC Nomenclature & Werner's Coordination Theory	$\text{Primary valency (ionizable, satisfies charge)}$, $\text{Secondary valency (non-ionizable, directional, satisfies CN)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Crystal Field Theory (CFT) & Isomerism (Very High Yield — 176 PYQs)

Core Principle: d-orbital splitting under octahedral and tetrahedral fields, high-spin vs low-spin complexes, spectrochemical series, and optical/geometrical isomerism.

■ **Caution / JEE Pitfall:** Tetrahedral complexes NEVER form low-spin complexes because $\Delta_t = (4/9)\Delta_o < P$ (splitting is always less than pairing energy).

■ **Speed Trick:** cis-[Co(en)₂Cl₂]⁺ is optically ACTIVE (non-superimposable mirror image), whereas the trans isomer has a plane of symmetry and is optically INACTIVE.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Which of the following sequences of hybridisation, geometry and magnetic nature are correct for the given coordination compounds? A. $[NiCl_4]^{2-}$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Match the LIST-I with LIST-II List-I Complex ion List-II Calculated spin only magnetic moment (BM) A. $[Cr(H_2O)_6]^{2+}$ 1. 3.0" [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "The correct increasing order of spin-only magnetic moment values of the complex ions $[MnBr_4]^{2-}$ (A), $[Cu(H_2O)_6]^{2+}$ (B), $[Ni(CN)_4]^{2-}$..." [Ans: N/A]

3.2 IUPAC Nomenclature & Werner's Coordination Theory (Very High Yield — 164 PYQs)

Core Principle: Formulation of complexes, Werner's postulates, precipitable silver chloride fractions, and systematic IUPAC naming conventions.

■ **Caution / JEE Pitfall:** In $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$, adding excess AgNO_3 precipitates ONLY 2 moles of AgCl per mole of complex, because one Cl^- is coordinated.

■ **Speed Trick:** Chelating ligands (e.g. en, EDTA) form 5- or 6-membered rings with the central metal ion, imparting exceptional thermodynamic stability (chelate effect).

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Identify the correct statements from the following A. $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ is the most stable complex among $[\text{Fe}(\text{OH})_6]^{3-}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$." **[Ans: N/A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "Given below are two statements : Statement I : $[\text{CoBr}_4]^{2-}$ ion will absorb light of lower energy than $[\text{CoCl}_4]^{2-}$ ion. Statement..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#2 (MCQ):** "The correct statements about metal carbonyls are A. The metal-carbon bonds in metal carbonyls possess both σ and π character. B. ..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

Given below are two statements : Statement I : Crystal Field Stabilization Energy (CFSE) of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ is greater than that of $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$. Statement II : Potassium ferricyanide has a greater spin-only magnetic moment than sodium ferrocyanide. In the light of the above statements, choose the correct answer from the options given below :

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are true **[Correct]**
- (C) Both Statement I and Statement II are false
- (D) Statement I is false but Statement II is true

Correct Answer: B

Solution Analysis: Statement I: In $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$, Cr^{2+} is d^4 and with H_2O (weak field) it is high spin : $t_{2g}^3 e_g^1$. $\text{CFSE} = 3(-0.4\Delta_o) + 1(+0.6\Delta_o) = -0.6\Delta_o$ In $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$, Mn^{2+} is d^5 high spin: $t_{2g}^3 e_g^2$. $\text{CFSE} = 3(-0.4\Delta_o) + 2(+0.6\Delta_o) = 0$ So CFSE of the chromium complex is greater (more stabilizing). Statement I is true. Statement II: $\text{K}_3[\text{Fe}(\text{CN})_6]$: Fe^{3+} is d^5 ; CN^- is strong field $\rightarrow$ low spin t_{2g}^5 , 1 unpaired electron $\rightarrow \mu_{\text{so}} = \sqrt{1(1+2)} = \sqrt{3}$ BM. $\text{Na}_4[\text{Fe}(\text{CN})_6]$

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Morning Shift (Q#2)

Consider a mixture 'X' which is made by dissolving 0.4 mol of $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and 0.4 mol of $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ in water to make 4 L of solution. When 2 L of mixture 'X' is allowed to react with excess of AgNO_3 , it forms precipitate 'Y'. The rest 2 L of mixture 'X' reacts with excess BaCl_2 to form precipitate 'Z'. Which of the following statements is CORRECT?

- (A) 0.1 mol of 'Y' is formed.
- (B) 0.4 mol of 'Z' is formed.
- (C) 0.2 mol of 'Z' is formed. **[Correct]**
- (D) 'Y' is BaSO_4 and 'Z' is AgBr .

Correct Answer: C

Solution Analysis: For coordination compounds, only the ions outside the coordination sphere dissociate in water and give precipitates. 1) Identify free ions in solution (i) $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ Here Br^- is outside the bracket, so in water: $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br} \rightarrow [\text{Co}(\text{NH}_3)_5\text{SO}_4]^{++} + \text{Br}^-$ So it gives 1 mol Br^- per mol of salt. (ii) $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ Here SO_4^{2-} is outside the bracket, so: $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4 \rightarrow [\text{Co}(\text{NH}_3)_5\text{Br}]^{2+} + \text{SO}_4^{2-}$ So it gives 1 mol SO_4^{2-} per mol of salt. 2) Moles present in 2 L of mixture X Total